

Exercises: Constrained Optimization

Exercise 1 (Penalization). *We consider the following problem*

$$(P) \quad \min_{x \in \mathbb{R}^n} \quad f(x) \\ \text{s.t.} \quad Ax = b, \quad x \leq 0$$

with value v and the following penalized versions

$$(P_t^{in}) \quad \min_{x \in \mathbb{R}^n} f(x) - t \sum_{i=1}^n \ln(-x_i) \\ \text{s.t.} \quad Ax = b, \quad x < 0$$

and

$$(P_t^{out}) \quad \min_{x \in \mathbb{R}^n} f(x) + t \sum_{i=1}^n (x_i)^+ \\ \text{s.t.} \quad Ax = b$$

with associated value v_t^{in} and v_t^{out} , and an optimal solution x_t^{in} and x_t^{out} .

1. Intuitively, assuming that f is "well behaved", for t going to which value does (P_t^{in}) tend to the original problem (P) ? In which sense?
2. What can you say about x_t^{in} ?
3. Can you compare v_t^{in} and v ?
4. Same questions for (P_t^{out}) .

Answers: For t going to 0 we have that (P_t^{in}) tends toward (P) : under suitable strict feasibility and compactness/coercivity assumptions, any accumulation point of x_t^{in} is an optimal solution of (P) , and $f(x_t^{in}) \rightarrow v$. In any case x_t^{in} is strictly admissible: $Ax_t^{in} = b$ and $x_t^{in} < 0$. Thus $f(x_t^{in}) \geq v$. There is no general comparison between v_t^{in} and v , since the barrier term $-t \sum_i \ln(-x_i)$ may be positive or negative.

For t going to $+\infty$, we have that (P_t^{out}) tends toward (P) : under suitable compactness/coercivity assumptions, the constraint violation $\sum_i (x_{t,i}^{out})^+$ goes to 0, any accumulation point of x_t^{out} is an optimal solution of (P) , and $v_t^{out} \rightarrow v$. The point x_t^{out} need not be admissible for finite t . We always have $v_t^{out} \leq v$, because every admissible solution of (P) has zero penalty. If the problem is convex and the constraints are qualified, this L^1 penalty is exact: for t large enough, an optimal solution of (P_t^{out}) is also an optimal solution of (P) .

Exercise 2 (A naive feasible direction). *Let $X = \mathbb{R}_+^2$ and $f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 1)^2$. Consider the following "masked gradient" direction:*

$$d_i(x) = -\nabla_i f(x) \mathbb{1}_{x_i > 0}.$$

1. Is $d(x)$ a feasible descent direction?
2. Apply the corresponding algorithm from $x^{(0)} = (0, 1)$ and comment the result.

Answers:

1. The direction is feasible for $X = \mathbb{R}_+^2$: if $x_i = 0$ then $d_i(x) = 0$, and if $x_i > 0$ then $x_i + td_i(x) \geq 0$ for all $t > 0$ small enough. It is a descent direction when it is nonzero, since

$$\nabla f(x)^\top d(x) = - \sum_{i: x_i > 0} (\nabla_i f(x))^2 < 0.$$

However, the direction may be zero even at a non-optimal point.

2. At $x^{(0)} = (0, 1)$, we have $\nabla f(x^{(0)}) = (-2, 0)$, hence $d(x^{(0)}) = (0, 0)$. The algorithm stops immediately at $(0, 1)$. This point is not optimal, since the unconstrained minimizer $(1, 1)$ belongs to X and has a smaller value. The issue is that freezing boundary coordinates can prevent movement into the feasible set, even when such a movement would decrease the objective.

Exercise 3 (Decomposition by prices). We consider the following energy problem:

- you are an energy producer with N production units
- you have to satisfy a given demand profile for the next 24h (i.e. the total output at time t should be equal to d_t)
- the time step is the hour, and each unit has a production cost over the horizon given as a convex quadratic function of the production profile
- For each unit i , the production planning $u^i = (u_t^i)_{t \in [24]}$ has to satisfy polyhedral constraints $u^i \in U^i$.

1. Model this problem as an optimization problem. To which class does it belong? How many variables?
2. Apply Uzawa's algorithm to this problem. Why could this be an interesting idea?
3. Give an economic interpretation to this method.
4. What would happen if the production units had coupling constraints between them? For example, if at any time step the sum of the production of units 1 and 2 could not exceed a given amount M_t .
5. What would happen if each unit had start-up costs?

Answers:

1. Let $u^i \in \mathbb{R}^{24}$ be the production profile of unit i , and let $C_i(u^i)$ be its convex quadratic production cost. The problem can be written as

$$\begin{aligned} \min_{u^1, \dots, u^N} \quad & \sum_{i=1}^N C_i(u^i) \\ \text{s.t.} \quad & u^i \in U^i, \quad i \in [N], \\ & \sum_{i=1}^N u_t^i = d_t, \quad t \in [24]. \end{aligned}$$

This is a convex quadratic program with $24N$ variables, since the objective is convex quadratic and the constraints are polyhedral and linear.

2. Dualizing the demand constraints with multipliers $\lambda = (\lambda_t)_{t \in [24]}$ gives the Lagrangian

$$L(u, \lambda) = \sum_{i=1}^N C_i(u^i) + \sum_{t=1}^{24} \lambda_t \left(\sum_{i=1}^N u_t^i - d_t \right).$$

For a fixed λ , the minimization over u decomposes by production unit:

$$u^i(\lambda) \in \arg \min_{u^i \in U^i} \left\{ C_i(u^i) + \sum_{t=1}^{24} \lambda_t u_t^i \right\}.$$

Uzawa's algorithm then updates prices by a gradient ascent step on the dual function, for instance

$$\lambda_t^{(k+1)} = \lambda_t^{(k)} + \rho_k \left(\sum_{i=1}^N u_t^i(\lambda^{(k)}) - d_t \right), \quad t \in [24].$$

This is interesting because the large centralized QP is replaced by N independent unit-level QPs plus a price update.

3. The economic interpretation is that λ_t is the electricity price at time t (up to the sign convention chosen in the Lagrangian). Given this price, each unit chooses its production profile independently: it produces more at time t only if the price compensates its marginal production cost and its own technical constraints. The role of the price update is to coordinate the independent units. If total production is too low compared with demand at time t , the price signal should encourage more production at that time; if total production is too high, it should discourage production. At convergence, the prices are such that the independent decisions of all units exactly satisfy the demand constraints.

4. If there are coupling constraints between units, such as

$$u_t^1 + u_t^2 \leq M_t, \quad t \in [24],$$

then the decomposition by individual units is no longer immediate: units 1 and 2 appear in the same constraint. One can either solve a joint subproblem for the coupled units, or dualize these coupling constraints as well, introducing additional prices for the shared resource.

5. Start-up costs make the model more difficult because they depend on whether a unit is switched on, and often on changes between consecutive time steps. A standard model introduces binary variables for the on/off status and start-up decisions. The problem then becomes a mixed-integer quadratic program rather than a convex QP. Price decomposition may still be used heuristically or inside a Lagrangian relaxation, but convexity and strong duality are no longer guaranteed, and a duality gap may appear.

Exercise 4 (Kelley's convergence). *We are going to prove that, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, and X a non-empty polytope (bounded polyhedron), then Kelley's cutting plane algorithm converges.*

Consider $x^{(1)} \in X$. Given $x^{(1)}, \dots, x^{(k)}$, with $g^{(\kappa)} \in \partial f(x^{(\kappa)})$ for all $\kappa \in [k]$, define $(x^{(k+1)}, \underline{v}^{(k+1)})$ as an optimal solution to $\mathcal{P}^{(k)}$:

$$\begin{aligned} \underline{v}^{(k+1)} &= \min_{x \in X, z \in \mathbb{R}} z \\ \text{s.t. } & f(x^{(\kappa)}) + \langle g^{(\kappa)}, x - x^{(\kappa)} \rangle \leq z, \\ & \kappa \in [k]. \end{aligned}$$

and then choose $g^{(k+1)} \in \partial f(x^{(k+1)})$.

Denote $v = \min_{x \in X} f(x)$.

1. *Show that v exists and is finite, and that there exists such a sequence $(x^{(k)})_{k \in \mathbb{N}^*}$.*
2. *Show that there exists L such that, for all k_1 and k_2 , we have $|f(x^{(k_1)}) - f(x^{(k_2)})| \leq L\|x^{(k_1)} - x^{(k_2)}\|$, and $\|g^{(k)}\| \leq L$.*
3. *Let $K_\varepsilon = \{k \in \mathbb{N}^* \mid f(x^{(k)}) > v + \varepsilon\}$ be the set of indices such that $x^{(k)}$ is not an ε -optimal solution. Show that $f(x^{(k)}) \rightarrow v$ if and only if K_ε is finite for all $\varepsilon > 0$.*

4. *Consider $k_1, k_2 \in K_\varepsilon$, such that $k_2 > k_1$. Show that*

$$f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle \leq \underline{v}^{(k_2)} \leq v$$

5. *Show that $\varepsilon + f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle < f(x^{(k_2)})$*

6. *Show that $\varepsilon < 2L\|x^{(k_2)} - x^{(k_1)}\|$.*

7. *Prove that $f(x^{(k)}) \rightarrow v$.*

8. *(Optional - hard) Deduce a complexity bound in terms of the maximum number of points in X that are pairwise at distance at least $\varepsilon/(2L)$.*

Answers:

1. f is finite convex and thus continuous on X which is compact, yielding the existence and finiteness of v .

f is subdifferentiable on $\mathbb{R}^n$, thus we have the existence of each $g^{(k)}$. Moreover, $\mathcal{P}^{(k)}$ admits an optimal solution: since X is compact and the maximum of finitely many affine cuts is continuous on X , this is equivalent to minimizing a continuous function on X .

2. We have seen that on any compact K included in the domain of a convex function f , f is L -Lipschitz. Here $\text{dom}(f) = \mathbb{R}^n$, so on a compact neighborhood of X , f is L -Lipschitz, and on X any subgradient g is of norm lower than L .

3. $f(x^{(k)}) \rightarrow v$ iff $\forall \varepsilon > 0, \exists N_\varepsilon \in \mathbb{N}, k \geq N_\varepsilon \implies f(x^{(k)}) \leq v + \varepsilon$. This is equivalent to $K_\varepsilon \subset [N_\varepsilon]$, hence to K_ε being finite.

4. By the subgradient inequality, for any $\kappa \leq k$ and any $x \in X$,

$$f(x) \geq f(x^{(\kappa)}) + \langle g^{(\kappa)}, x - x^{(\kappa)} \rangle.$$

Hence the cutting-plane model underestimates f on X , and therefore $\underline{v}^{(k)} \leq v$ for all k . Moreover, since $(x^{(k_2)}, \underline{v}^{(k_2)})$ is feasible for $\mathcal{P}^{(k_2-1)}$, it satisfies in particular the cut $\kappa = k_1$, giving

$$f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle \leq \underline{v}^{(k_2)} \leq v.$$

5. As $k_2 \in K_\varepsilon$, we have $f(x^{(k_2)}) > v + \varepsilon \geq f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle + \varepsilon$ by the previous question.

6. We have

$$\begin{aligned} \varepsilon &< |f(x^{(k_2)}) - f(x^{(k_1)})| + |\langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle| \\ &\leq 2L \|x^{(k_2)} - x^{(k_1)}\| \end{aligned}$$

by Cauchy-Schwarz and question 2.

7. If $f(x^{(k)}) \not\rightarrow v$, then there exists $\varepsilon > 0$ such that K_ε is infinite. As X is compact, we can extract a converging subsequence $(x^{(\sigma(j))})_{j \in \mathbb{N}}$ with $\sigma(j) \in K_\varepsilon$. Therefore there exist $j_2 > j_1$ such that

$$\|x^{(\sigma(j_2))} - x^{(\sigma(j_1))}\| < \frac{\varepsilon}{2L}.$$

This contradicts question 6, applied with $k_1 = \sigma(j_1)$ and $k_2 = \sigma(j_2)$.

8. Question 6 shows that all indices in K_ε correspond to points of X that are pairwise at distance at least $\varepsilon/(2L)$. Therefore $|K_\varepsilon|$ is bounded by the packing number

$$N\left(X, \frac{\varepsilon}{2L}\right),$$

the maximum cardinality of a subset of X with pairwise distances at least $\varepsilon/(2L)$. After more than this number of iterations, at least one iterate must be ε -optimal. For instance, if X is contained in a ball of radius R , this gives a rough bound of order $(LR/\varepsilon)^n$.